COGNITION: DISTRIBUTED DATA PROCESSING SYSTEM FOR LUNAR ACTIVITIES

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ABSTRACT

Moon exploration has gained significant momentum in recent decades, with growing interest from space agencies and private investors. A diverse range of activities is associated with moon exploration, encompassing spacecraft design, payload transportation, launcher capabilities, resource identification and mining, and establishing a sustained presence on our only natural satellite. Such endeavors would require the shipment of both scientific and life-sustaining equipment. However, communication between Earth's mission control and the Moon's bases can still present challenges. In this paper, we introduce Cognition – a rover-lander distributed system that approaches this problem by distributing the data processing between the rover and the lander. The primary goal of the Cognition system is to optimize lunar surface exploration by minimizing data transmission to the Earth's surface, prioritizing the transfer of valuable data, and augmenting the level of autonomy in the process.

***Index Terms*—** Robotics, Space, Remote sensing, ROS

1. INTRODUCTION

Moon exploration has only started within the last few decades, and the interest continues to grow. Space agencies and private investors increasingly recognize the vast opportunities that moon exploration presents and are expressing greater commercial interest in this field. This growing enthusiasm drives investments and collaborations to unlock the economic and scientific potential of lunar exploration. According to a Euroconsult study, 60 missions to the Moon are planned in 2023-2031 [1], 41 being commercial ones, allowing to explore Earth's natural satellite better and prepare for long-term Moon exploration and, in the future, deep space. The lunar exploration market (both governmental and commercial entities) is expected to grow from $2.2 billion in 2022 to $5 billion by 2031. Lunar landers and rovers' development will be essential in exploration activities. The primary mission of lunar landers and rovers is to collect, aggregate, and process a growing volume of complex data. This includes various types of data, such as vision data obtained from cameras, radar data, LIDAR data, scientific payload data, and telemetry data. These vehicles serve as crucial platforms for gathering and analyzing diverse datasets to enhance our understanding of the lunar environment[2, 3]. The distance to the Moon impedes data transmission to the Earth's surface, forcing it to move at least part of data processing to its surface.

In our solution, the onboard computer and the data processing unit (DPU) mounted on the rover are responsible for queuing and initial processing of collected data according to the edge processing paradigm [4]. Lander's more powerful DPU, built of redundant accelerators with higher computing power, will be responsible for complex data processing and storage. The primary motivation behind Cognition is to limit the transfer of data to the Earth's surface, ensure that only valuable data is being sent to the ground, and introduce more autonomy into lunar surface exploration. Such design brings an additional advantage – docking the rover at the lander facilitates the survival of the moon night [5]. This ensures that the rover can leverage the resources and support systems of the lander during the extended periods of darkness on the moon's surface. The conceptual diagram of the architecture is given in Fig. 1.

The presented study is the first step toward implementing the Cognition concept. In our work, we focus on four objectives contributing to the study's primary goal, namely: (*i*) design of the proposed distributed architecture, (*ii*) thorough benchmarking of the base computational platforms used as the heart of rover and lander DPUs, (*iii*) integration of the parts of the system (sensors, DPU, rover base) within a single, extendable software architecture Robot Operating System (ROS) and finally (*iv*) – demonstrating the operation of the integrated rover in an analog environment. These objectives set the foundation for the Cognition system's successful

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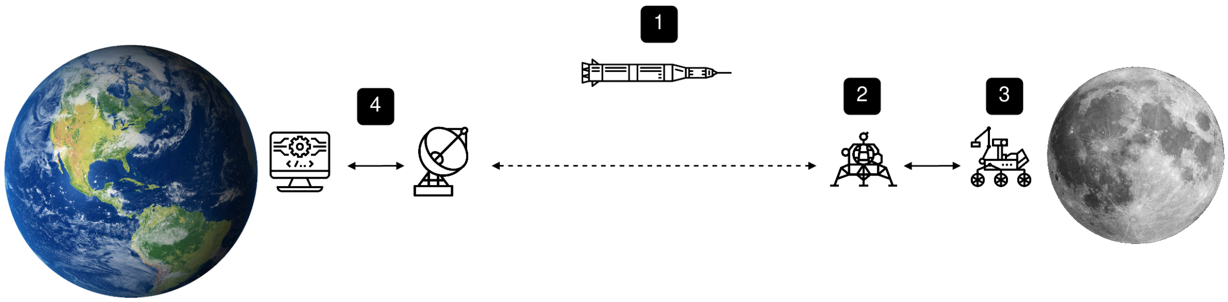

Fig. 1. Cognition rover-lander distributed system proposed architecture consisting of (1) rocket delivery system operated by a subcontractor, (2) moon lander with long-range communication module, (3) rover for exploration exchanging data only with the lander and (4) a base operation station on Earth.

implementation.

2. SYSTEM OVERVIEW

2.1. Hardware description

We show the capabilities of the artificial intelligence (AI) development environment from Xilinx (Vitis AI) [6] and evaluate two hardware architectures (Leopard DPU, Versal AI) that could be used in future lunar missions. We implemented resource-frugal AI algorithms for object detection and segmentation (detecting large rocks, the most common obstacles in lunar rover maneuvering). We deployed them on two DPU platforms with Vitis AI – a tool that is expected to support seamless AI models transfer to edge devices. We also performed a thorough benchmark to examine the computing units' performance and power budgets. The final benchmark was performed using the dataset with lunar surface images (we exploited the Artificial Lunar Landscape Dataset [7] for this purpose) and implementing an algorithm that could denote the presence of specific objects in these images. Further, we performed DPU platforms benchmarking: deploying an object detection algorithm and comparing the processing efficiency of two selected DPUs. The results reveal that the Versal-based solution is notably faster and achieves about ten times better power efficiency than previous-generation devices. Subsequently, we seamlessly incorporated our resource-frugal models, designed for the Xilinx platform, into the mobile robot system. This integration involved leveraging hybrid, system-on-chip architectures that combine conventional processors with programmable hardware. These promising hardware configurations hold great potential for space robotics applications. By capitalizing on the advantages offered by these hybrid architectures, we enhance the performance and adaptability of our mobile robot system, positioning it for optimized operation in space exploration scenarios.

2.2. Software description

To leverage the full potential stemming from the hardware platform's capabilities and hybrid nature, we deployed the Linux base operating system and the ROS2 distribution on the integrated CPU of the benchmarked DPUs. The rover's software is based on Robot Operating System (ROS2) [8]. ROS2 is a highly modular environment, and many open-source packages developed by hobbyists, scientists, and industry are available, yet it's run chiefly using typical processors. Our preference for ROS2 is reinforced by NASA's choice of robot software for their future mission [9], called Space ROS. Our study demonstrated the feasibility of utilizing pre-existing ROS software to a significant extent. We explored the potential for future developments in Space ROS. We highlighted the need for dedicated efforts to develop custom packages tailored for limited on-board, hybrid system-on-chip processors that seamlessly integrate processors and programmable hardware components. The Space ROS pushes the envelope further and underlines the requirements for flight-quality robotic and autonomous space systems.

In the final phase, we conducted analog tests that involved the utilization of a DPU connected to a stereovision camera. The output data stream from the camera was processed using a solution that underwent benchmarking during the initial stages of the project. This allowed us to perform final unit and integration tests by driving the rover in an unknown environment close to the target operating conditions. The block diagram of the implemented architecture and the envisioned future work is given in Fig. 2.

We also provided the chart regarding the tests of the DPU power consumption under the workloads (neural network inference) developed in our project. It is shown in Figure 3.

3. DISCUSSION AND FUTURE WORK

By directly working on the study outcomes, we reported successfully validating the proposed system and its sub-systems (sub-goals), establishing their potential as a feasible solution

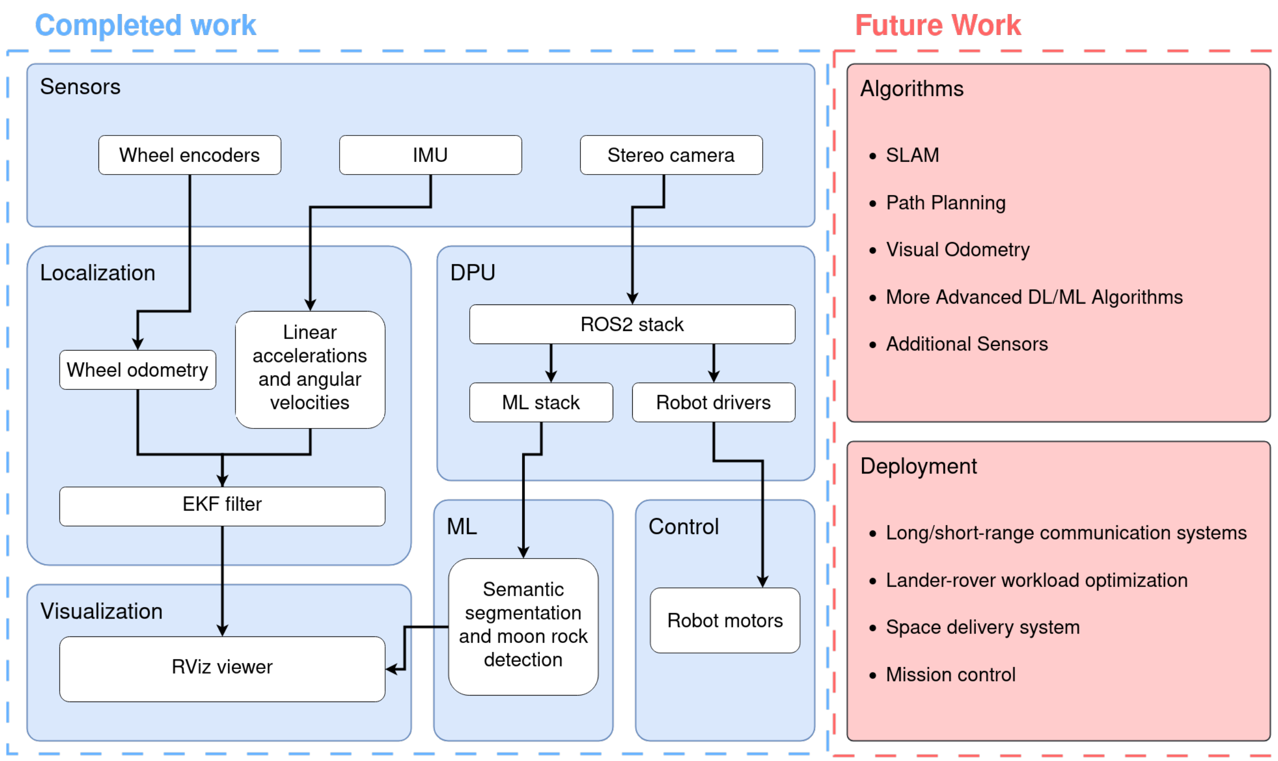

Fig. 2. Summary of our work done compared to the future possible developments in the Cognition framework.

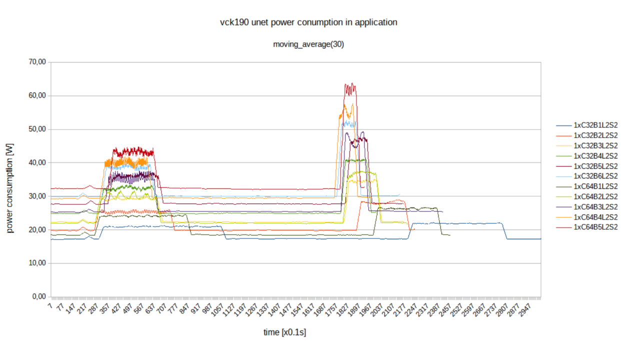

Fig. 3. Power consumption of the proposed hardware architectures in the network inference process.

for lunar missions. The deliberate construction of a distributed architecture, as demonstrated in our experiments, enabled us to delegate non-real-time computational tasks to a lander's processing unit, significantly increasing the overall computational power of a rover mission. This proof-of-concept solution allowed us to deploy AI algorithms onboard a mobile robotic platform and provided valuable insights and expertise in this field. The proof-of-concept solution for deploying AI algorithms onboard a mobile robotic platform, testing already existing components, gaining additional expertise in the field, and creating a path-to-fly for a distributed system for rovers and landers applications is a significant step towards the realization of the complete Cognition concept.

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